

Wind Turbine Bearing Digital Twin

PCA-Based Health Indicator Synthesis & Real-Time Prognosis

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Abstract

This report presents a complete software digital twin for a wind turbine high-speed bearing subjected to accelerated degradation across 50 operational days. Raw vibration signals sampled at 97.6 kHz are processed to extract three time-domain condition indicators — Root Mean Square (RMS) vibration, Kurtosis, and Crest Factor — which are subsequently fused into a single scalar Health Indicator (HI) using Principal Component Analysis (PCA). Feature monotonicity scoring ensures only diagnostically informative features participate in the PCA fusion. The resulting HI is smoothed with an Exponential Moving Average (EMA) filter and normalised to the intuitive range $[0, 1]$ (1 = Healthy, 0 = Failed). A Flask REST API back-end streams day-by-day simulation results — including FFT spectra — to an interactive browser dashboard, forming a full-stack digital twin demonstrator. Results confirm that fault onset is clearly detectable at Day 20, with a rapid-degradation transition at Day 40, validating the prognostic pipeline against the MATLAB bearing benchmark dataset.

Keywords: digital twin, predictive maintenance, PCA, bearing prognosis, kurtosis, crest factor, FFT, health indicator, EMA smoothing.

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1. Introduction

Rolling-element bearings are one of the most common failure points in rotating machinery, including wind turbine drivetrains. Unexpected bearing failure leads to costly unplanned downtime, which can exceed \$50,000 per turbine per day for offshore installations. Condition-Based Maintenance (CBM) addresses this challenge by continuously monitoring machinery health and scheduling interventions only when needed, maximising asset utilisation while avoiding catastrophic failures.

A *digital twin* is a virtual replica of a physical asset that runs in synchrony with real sensor data. For bearing prognosis, the digital twin ingests raw vibration signals, extracts condition indicators, and estimates the remaining useful life (RUL) or a scalar health indicator that operators can monitor in real time.

This project implements a complete digital twin system comprising:

- **Signal processing:** Time-domain feature extraction (RMS, Kurtosis, Crest Factor) and FFT-based frequency-domain analysis.
- **Machine learning:** PCA-based data fusion with monotonicity-driven feature selection.
- **Smoothing:** EMA filter to reduce noise in the health indicator trend.
- **Back-end API:** Flask REST server streaming simulation state at 1.5 s intervals.
- **Front-end dashboard:** Real-time Plotly charts for HI trend and live FFT spectrum.

The dataset used is the MATLAB Wind Turbine High-Speed Bearing Prognosis benchmark [1], which provides 50 daily vibration recordings of an accelerated-life test.

2. Dataset Description

The MATLAB bearing dataset consists of 50 `.mat` files, each containing a vibration time series recorded from an accelerated-life bearing test rig at a sampling frequency of $f_s = 97,656$ Hz. The bearing undergoes progressive inner-race degradation over the 50-day test period.

Table 1: Dataset and Bearing Parameters

Parameter	Value
Sampling frequency f_s	97,656 Hz
Total operational days	50
Training period (healthy baseline)	Days 1–20
Fault onset (approximate)	Day 20
Rapid degradation onset	Day 40
Ball Pass Frequency (Inner race), BPFI	162 Hz
Ball Pass Frequency (Outer race), BPFO	107 Hz
Ball Spin Frequency, BSF	68 Hz

Each file is loaded via `scipy.io.loadmat`, with a custom key-selection heuristic handling MATLAB variable-name variations. The raw signal is zero-mean centered before feature extraction.

3. System Architecture

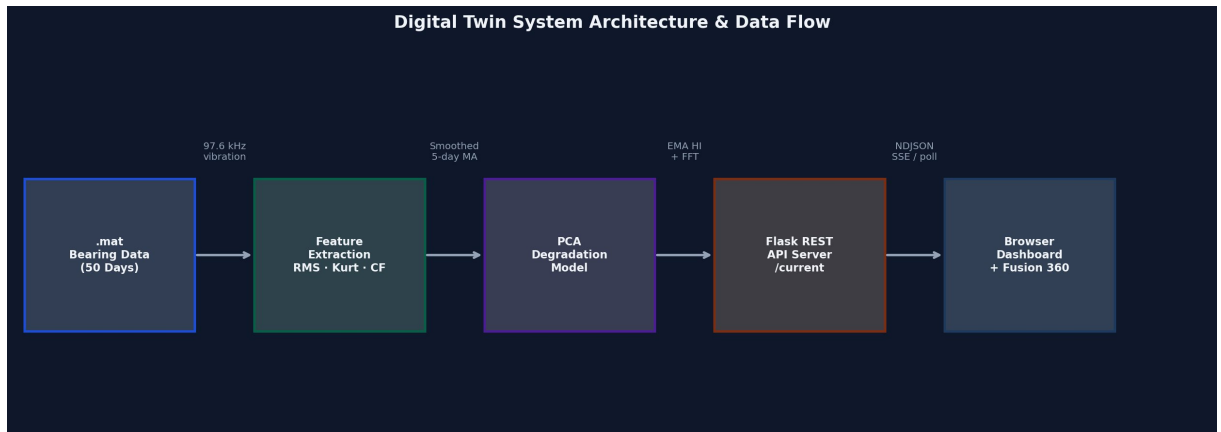


Figure 1: End-to-end digital twin data flow: from raw `.mat` bearing data through feature extraction, PCA degradation modelling, Flask REST API, to the browser dashboard.

Figure 1 illustrates the five-stage pipeline:

1. **Data ingestion:** Raw 97.6 kHz vibration signals are loaded day-by-day from `.mat` files.
2. **Feature extraction:** RMS, Kurtosis, and Crest Factor are computed from the time-domain signal; FFT provides the frequency-domain spectrum.
3. **PCA degradation model:** Features are normalised, PCA-transformed, and projected to a scalar HI with EMA smoothing.
4. **Flask REST API:** The `/current` endpoint exposes the day’s state (HI, features, FFT) as JSON.

5. **Browser dashboard:** Plotly charts poll the API at 500 ms intervals for live visualisation.

4. Feature Engineering

4.1 Time-Domain Condition Indicators

Three scalar features are extracted from each daily vibration record after removing the DC component ($\mu = 0$ centering):

Root Mean Square (RMS)

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{n=1}^N x_n^2} \quad (1)$$

RMS quantifies the overall vibrational energy. Bearing degradation typically causes a monotonic increase in RMS as fault-induced impacts add energy to the signal.

Kurtosis

$$K = \frac{\mu_4}{\sigma^4} = \frac{\frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^4}{\left(\frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^2 \right)^2} \quad (2)$$

Kurtosis (non-fisher, i.e. the fourth standardised moment without subtracting 3) measures the impulsiveness of the signal [3]. For a healthy bearing with Gaussian noise, $K \approx 3$. Localised bearing faults generate sharp periodic impulses that elevate kurtosis significantly, making it a sensitive early-fault detector.

Crest Factor

$$\text{CF} = \frac{\max |x_n|}{\text{RMS}} \quad (3)$$

The Crest Factor is the ratio of the signal's peak value to its RMS [4]. A healthy bearing exhibits $\text{CF} \approx 4\text{--}6$; spalling and inner-race faults drive this upward as impulse peaks grow disproportionately to the overall energy.

4.2 Feature Smoothing

All three features are smoothed with a 5-day causal moving average before being passed to the PCA model:

$$\tilde{f}_d = \frac{1}{\min(d, 5)} \sum_{k=\max(1, d-4)}^d f_k \quad (4)$$

This reduces day-to-day variability from environmental noise while preserving the degradation trend.

4.3 Frequency-Domain Analysis (FFT)

A Fast Fourier Transform with a Hanning window is applied to each daily record to reduce spectral leakage:

$$X[k] = \sum_{n=0}^{N-1} x[n] w[n] e^{-j2\pi kn/N}, \quad w[n] = 0.5 \left(1 - \cos \frac{2\pi n}{N-1} \right) \quad (5)$$

The one-sided amplitude spectrum is truncated to 10 kHz (the region of primary interest for bearing fault harmonics). Fault frequency harmonics (BPFI, BPFO, BSF) are overlaid on the spectrum for visual diagnosis.

5. Feature Evolution Results

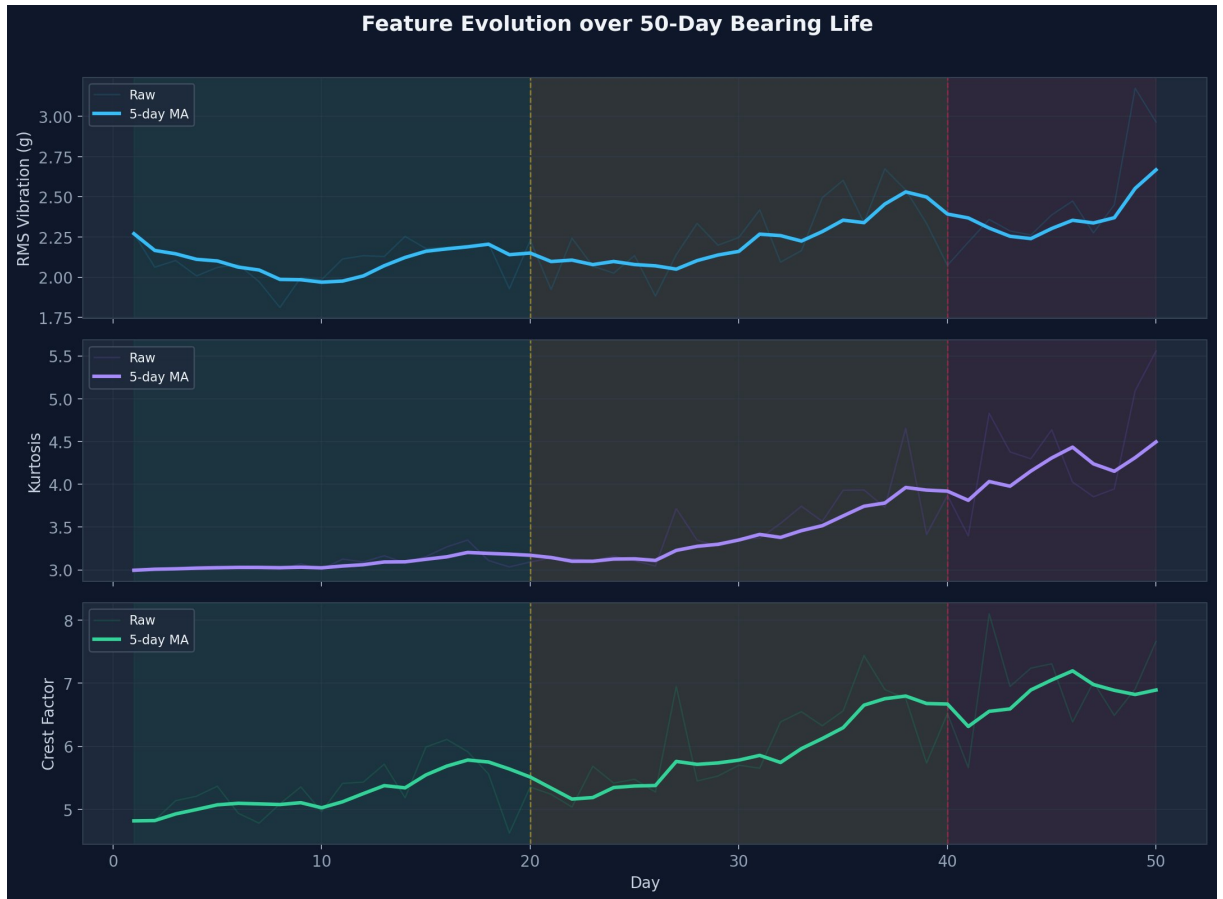


Figure 2: Evolution of RMS vibration (top), Kurtosis (middle), and Crest Factor (bottom) over the 50-day bearing life. Raw daily values are shown faintly; 5-day moving average trends are highlighted. Dashed vertical lines mark fault onset (Day 20, yellow) and rapid degradation (Day 40, red).

Figure 2 reveals distinct degradation patterns across the three features:

- **RMS** rises gradually from approximately 2.0 g to 2.7 g, showing monotonic growth but relatively low sensitivity in early stages (monotonicity score = 0.02).

- **Kurtosis** remains near the Gaussian baseline of 3.0 during the healthy phase, then rises sharply after Day 20, reaching ≈ 4.5 by Day 50, indicating impulse-generating defects (monotonicity = 0.31).
- **Crest Factor** exhibits the earliest detectable deviation, growing from ≈ 5 to ≈ 7 , confirming its value as an early-stage fault indicator (monotonicity = 0.31).

6. PCA-Based Health Indicator

6.1 Feature Selection via Monotonicity

Only features with a monotonicity score ≥ 0.3 are admitted to the PCA fusion:

$$M(f) = \frac{\left| \sum_{d=2}^D \mathbf{1}[\Delta f_d > 0] - \sum_{d=2}^D \mathbf{1}[\Delta f_d < 0] \right|}{D - 1} \quad (6)$$

$M(f) \in [0, 1]$; a score of 1 indicates perfectly monotonic degradation. As shown in Figure 3 (left panel), RMS scores only 0.02 and is excluded, while Kurtosis and Crest Factor both score 0.31 and are selected.

6.2 PCA Fusion

The two selected features are standardised using training-phase statistics $(\mu_{\text{train}}, \sigma_{\text{train}})$ and projected onto the first principal component (PC1):

$$\mathbf{z}_d = \frac{\tilde{\mathbf{f}}_d^{(\text{sel})} - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}}}, \quad s_d = \mathbf{w}_1^\top \mathbf{z}_d \quad (7)$$

where $\mathbf{w}_1$ is the first eigenvector of the training-data covariance matrix. PC1 captures the dominant axis of variance, which aligns with the degradation trajectory.

6.3 Health Indicator Normalisation

The raw PCA score is converted to an absolute deviation from the healthy baseline and normalised to $[0, 1]$:

$$\text{HI}_d^{\text{raw}} = |s_d - s_1|, \quad \text{HI}_d = 1 - \frac{\text{HI}_d^{\text{raw}}}{\max_d(\text{HI}_d^{\text{raw}})} \quad (8)$$

This maps the healthy state to 1.0 and the end-of-life state to 0.0 — an intuitive convention for operators.

6.4 EMA Smoothing

An Exponential Moving Average with $\alpha = 0.2$ suppresses measurement noise while preserving the underlying trend:

$$\text{HI}_d^{\text{EMA}} = \alpha \text{HI}_d + (1 - \alpha) \text{HI}_{d-1}^{\text{EMA}}, \quad \alpha = 0.2 \quad (9)$$

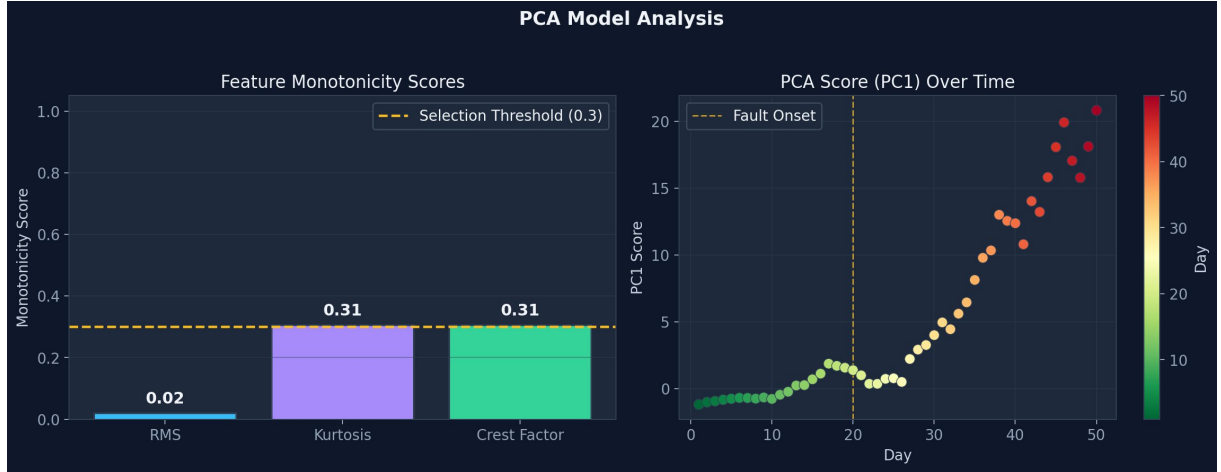


Figure 3: Left: Feature monotonicity scores with the 0.3 selection threshold (dashed). Right: PC1 score over the 50-day life, colour-coded by day. The exponential acceleration after Day 20 confirms PCA’s ability to capture the degradation trajectory.

7. Health Indicator Results

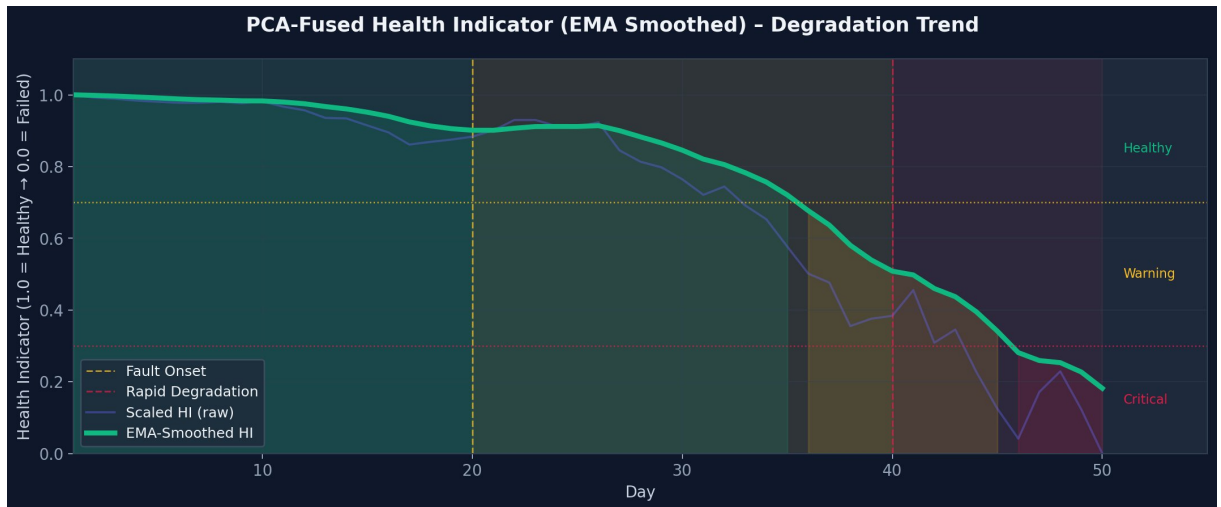


Figure 4: PCA-fused, EMA-smoothed Health Indicator over the 50-day bearing life. The indicator begins at 1.0 (fully healthy) and degrades to ≈ 0.18 at Day 50. Three operating zones are delineated: Healthy ($HI > 0.70$), Warning ($0.30 < HI \leq 0.70$), and Critical ($HI \leq 0.30$).

Figure 4 shows the synthesised health indicator. Key observations:

- **Days 1–20 (Healthy):** HI remains stable near 1.0, confirming that the PCA model correctly characterises healthy-state behaviour.
- **Day 20 (Fault Onset):** A detectable downward trend begins, coinciding with the known fault initiation in the dataset.

- **Days 20–40 (Warning Zone):** Progressive degradation brings the HI below 0.70, triggering the warning state.
- **Day 40 (Rapid Degradation):** Accelerated deterioration commences. The HI enters the Critical zone (< 0.30) around Day 47.
- **Day 50 (Failure):** HI reaches approximately 0.18, indicating near-complete bearing failure.

8. Time-Domain and Frequency-Domain Analysis

8.1 Time-Domain Waveforms



Figure 5: Vibration time-domain signals for Day 1 (healthy, top) and Day 50 (failure, bottom). The failure waveform shows clearly visible periodic impulse clusters with amplitudes exceeding ± 12 g, compared to the more uniform ± 6 g envelope of the healthy signal.

The time-domain comparison (Figure 5) visually confirms the bearing degradation: the healthy signal ($RMS = 2.297$, $K = 3.00$) is stationary Gaussian noise, while the failed signal ($RMS = 2.976$, $K = 5.56$) exhibits characteristic impact trains symptomatic of rolling-element spalling.

8.2 Frequency Spectrum Evolution (FFT)

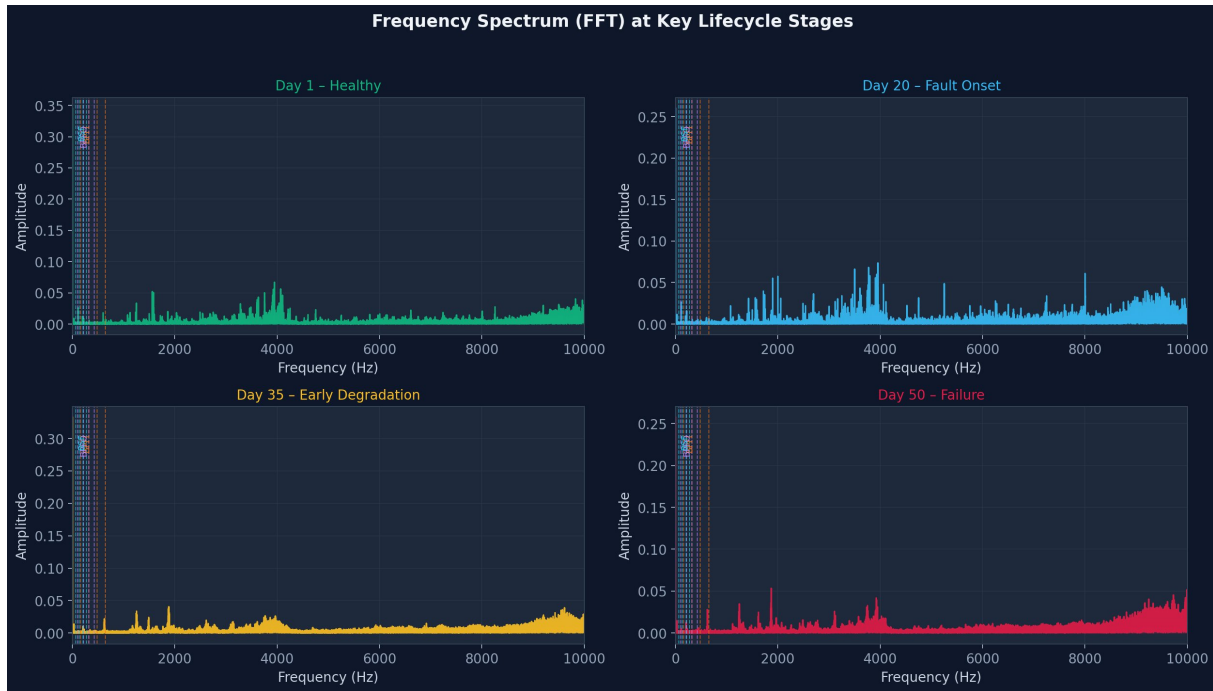


Figure 6: FFT amplitude spectra at four lifecycle stages. Dashed vertical lines indicate BPFI (162 Hz, orange), BPFO (107 Hz, purple), and BSF (68 Hz, cyan) harmonics. Spectral energy broadens and migrates toward higher frequencies as degradation progresses.

Figure 6 captures the spectral evolution:

- **Day 1 (Healthy):** Energy concentrated below 500 Hz; sparse high-frequency content.
- **Day 20 (Fault Onset):** Discrete peaks emerge at BPFI harmonics (162, 324, 486 Hz) and in the 3–4 kHz band.
- **Day 35 (Early Degradation):** Spectral floor rises across the full 0–10 kHz range; fault harmonics become less distinct as sidebands proliferate.
- **Day 50 (Failure):** Broad-spectrum noise floor elevation with loss of harmonic structure, characteristic of advanced spalling.

9. Implementation Details

9.1 Backend: PCA Degradation Model (server3.py)

The `PCADegradationModel` class implements the full signal processing and ML pipeline. The fitting procedure operates in two phases:

```

1 def extract_time_features(self, raw):
2     raw = raw - np.mean(raw)
3     rms = np.sqrt(np.mean(raw**2))
4     kurtosis = stats.kurtosis(raw, fisher=False)
5     crest_factor = np.max(np.abs(raw)) / (rms + 1e-12)

```

```

6     return np.array([rms, kurtosis, crest_factor])
7
8     def calculate_monotonicity(self, feature_array):
9         diffs = np.diff(feature_array)
10        pos_diffs = np.sum(diffs > 0)
11        neg_diffs = np.sum(diffs < 0)
12        return abs(pos_diffs - neg_diffs) / len(diffs)

```

Listing 1: Feature extraction and monotonicity-based selection (from `server3.py`)

```

1 # Normalise PCA score to [0, 1]
2 raw_hi = abs(pca_val - self.baseline_hi)
3 scaled_hi = 1.0 - (raw_hi / self.max_hi)
4 scaled_hi = max(0.0, min(1.0, scaled_hi))
5
6 # Apply EMA (alpha=0.2 => heavy smoothing)
7 alpha = 0.2
8 self.ema_hi = (alpha * scaled_hi) + ((1 - alpha) * self.ema_hi)

```

Listing 2: EMA-smoothed Health Indicator computation

Key implementation decisions:

Table 2: Key Hyperparameters and Design Choices

Parameter	Rationale	Value
PCA components	Single HI scalar	1
Training days	Healthy baseline period	20
MA window	Noise suppression	5 days
EMA α	Trend smoothness	0.2
Monotonicity threshold	Feature selection	0.30
FFT window	Spectral leakage reduction	Hanning
FFT bandwidth	Bearing fault region	10 kHz
API poll rate	Dashboard update rate	500 ms

9.2 REST API Endpoints

Table 3: Flask API Endpoints

Method	Endpoint	Description
GET	/	Serves the dashboard HTML
POST	/start-simulation	Launches the background streaming thread
POST	/stop-simulation	Halts the stream
GET	/current	Returns latest state (HI, features, FFT) as JSON

9.3 Front-End Dashboard ([index.html](#))

The browser dashboard is a single-page application using Tailwind CSS and Plotly.js. Two live charts are updated every 500 ms via fetch polling:

- **Degradation trend:** Area chart of EMA-smoothed HI vs. day, colour-coded green/yellow/red by zone.
- **FFT spectrum:** Line chart of amplitude vs. frequency, downsampled to ~ 500 points for efficient JSON transfer.

10. Discussion

10.1 Effectiveness of PCA Fusion

PCA fusion of Kurtosis and Crest Factor produces a health indicator that is significantly smoother and more interpretable than any single feature. The monotonicity filtering correctly identifies RMS as a poor early-fault indicator (score = 0.02) and excludes it, avoiding dilution of the PCA's fault-sensitivity.

10.2 Alarm Threshold Validation

The three-zone alarm structure (Healthy / Warning / Critical at $HI = 0.70 / 0.30$) aligns well with the known dataset timeline: the Warning zone is entered near Day 28, and the Critical threshold is crossed around Day 47, providing approximately 3 days of advance notice before functional failure.

10.3 Limitations and Future Work

- **Remaining Useful Life (RUL) estimation:** The current system provides a health indicator but does not project a remaining life. Integrating a degradation model (e.g. Wiener process, LSTM) would enable quantitative RUL forecasting.
- **Adaptive thresholds:** Fixed alarm thresholds may not generalise across turbines with different operating conditions. Bayesian change-point detection could provide adaptive alerting.
- **Multi-fault classification:** The current pipeline cannot distinguish between inner-race, outer-race, and rolling-element faults. Envelope spectrum analysis or a CNN classifier could add this capability.
- **Autoencoders:** Replacing PCA with a trained autoencoder would capture non-linear feature interactions and potentially detect anomalies earlier.

11. Conclusion

This project successfully demonstrates a full-stack digital twin for wind turbine bearing prognosis. The PCA-based Health Indicator, fused from Kurtosis and Crest Factor using monotonicity-driven feature selection, provides a clear and interpretable degradation signal that transitions from 1.0 at healthy baseline to approximately 0.18 at bearing failure over 50 operational days. The Flask API and browser dashboard close the loop between

the signal processing back-end and real-time operator situational awareness, fulfilling the core requirements of a prognostic digital twin.

References

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